

NEWQUAY CORNWALL, United Kingdom

WMO: 033465

Lat: 50.433N

Lon: 4.983W

Elev: 390

StdP: 14.49

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.8	33.1	23.4	18.0	34.5	26.0	20.2	36.1	37.3	48.5	34.1	46.3	9.9	70	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	73.5	63.1	70.2	61.7	67.6	60.7	65.2	70.3	63.6	67.6	62.4	65.7	11.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	88.8	66.2	62.1	84.5	64.9	61.0	81.2	63.8	30.2	70.5	29.1	67.8	28.2	65.7	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
32.2	28.8	25.7	DB	27.5	78.1	2.9	3.7	25.4	80.8	23.7	83.0	22.1	85.1	20.0	87.8
			WB	25.6	67.4	3.1	2.7	23.4	69.3	21.5	70.9	19.8	72.4	17.6	74.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	44.5	44.4	45.9	48.3	53.2	57.5	60.8	61.7	58.5	54.1	49.0	45.2
	DBStd	7.46	5.10	4.53	4.09	4.14	4.14	3.57	3.47	3.22	3.10	3.71	3.87	5.18
	HDD50	790	176	160	134	76	14	0	0	0	0	10	62	158
	HDD65	4773	636	577	591	500	364	228	139	114	195	337	479	613
	CDD50	1518	5	3	8	27	115	225	335	362	256	138	34	10
	CDD65	25	0	0	0	0	0	3	10	11	1	0	0	0
	CDH74	88	0	0	0	0	0	17	32	37	2	0	0	0
	CDH80	6	0	0	0	0	0	1	3	2	0	0	0	0
(14) Wind	WSAvg	12.6	15.3	14.6	13.6	12.1	11.4	10.9	10.3	10.3	11.3	13.5	13.2	14.7
(15) Precipitation	PrecAvg	44.3	4.8	4.1	3.0	3.0	2.7	2.6	2.8	2.9	3.1	4.8	5.2	5.2
	PrecMax	55.0	8.7	8.3	5.5	6.6	5.6	5.1	5.8	7.0	6.7	8.9	10.7	11.1
	PrecMin	35.2	1.1	0.9	0.8	0.6	0.3	0.7	0.3	0.6	0.9	2.0	2.2	1.4
	PrecStd	6.6	2.0	2.1	1.4	1.5	1.6	1.5	1.5	1.8	1.6	1.7	2.2	2.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	54.8	59.1	64.8	71.2	76.3	78.1	78.6	72.6	64.4	59.2	55.8
		MCWB	52.3	52.1	50.7	53.8	59.3	63.7	65.2	65.8	63.5	59.0	57.1	54.1
	2%	DB	53.7	53.1	54.9	59.9	66.9	69.8	73.2	72.7	68.1	61.8	57.0	54.7
		MCWB	52.0	51.1	50.1	52.4	57.1	60.2	63.0	64.3	61.2	58.1	55.1	53.3
	5%	DB	52.4	51.6	53.0	56.7	63.1	66.3	69.7	69.2	65.5	60.5	55.9	53.6
		MCWB	50.7	49.5	49.9	50.8	56.0	58.8	61.8	62.5	60.0	57.6	54.0	52.0
	10%	DB	51.0	50.4	51.4	54.2	59.7	63.6	66.6	66.7	63.7	59.4	54.8	52.3
		MCWB	49.3	48.4	49.1	49.9	54.4	57.7	60.6	61.7	59.1	57.0	52.8	50.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	53.6	53.1	56.1	61.0	65.1	66.4	67.7	65.2	60.9	57.9	54.9
		MCDB	54.0	54.4	54.6	60.5	67.9	74.4	75.1	73.2	69.8	62.2	58.6	55.2
	2%	WB	52.3	51.8	51.9	53.6	58.4	61.9	64.2	65.8	63.0	59.5	56.1	53.9
		MCDB	53.2	52.8	53.6	57.7	64.1	67.1	69.6	70.0	66.0	60.7	56.6	54.6
	5%	WB	51.0	50.3	50.8	52.1	56.8	60.1	62.9	64.2	61.6	58.5	54.7	52.6
		MCDB	52.1	51.1	52.3	55.3	61.4	64.1	67.0	67.5	64.0	59.8	55.5	53.4
	10%	WB	49.8	49.0	49.7	50.8	55.1	58.7	61.7	62.8	60.4	57.6	53.1	51.1
		MCDB	50.9	50.1	51.1	53.4	58.5	62.1	65.3	65.7	62.6	59.0	54.3	52.1
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.7	8.2	9.5	10.0	9.4	9.0	9.1	9.4	8.0	7.2	7.4
		MCDBR	7.4	7.7	10.1	12.9	15.2	14.2	14.7	12.8	12.1	8.6	7.1	7.1
	5% WB	MCWBR	7.5	7.1	7.0	7.8	8.9	8.2	7.7	6.8	7.2	6.6	7.0	8.1
		MCWBR	7.2	6.9	8.1	10.6	12.9	12.1	11.7	11.0	10.1	7.7	6.4	7.0
(40) Clear-Sky Solar Irradiance	taub		0.333	0.345	0.371	0.379	0.387	0.374	0.378	0.363	0.360	0.353	0.331	0.324
		taud	2.364	2.392	2.335	2.340	2.358	2.423	2.417	2.477	2.474	2.456	2.412	2.329
	Ebn at Noon		218	249	262	274	276	280	277	276	265	243	218	200
		Edh at Noon	21	26	33	36	37	35	35	31	28	24	20	19
(44) All-Sky Solar Radiation	RadAvg		254	507	869	1399	1696	1869	1707	1458	1097	620	325	204
	RadStd		24	49	94	93	114	167	120	97	73	42	27	23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page